AI Multi-Agent Supply Chain Control Tower

“Autonomous AI agents that detect, decide, and act in real-time supply chains.”

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PROBLEM STATEMENT-

Modern Supply Chains Are Reactive, Not Intelligent

- Delays are detected too late
- No real-time failure detection
- Manual decision-making slows response
- Lack of visibility across workflows
- No auditability of decisions

Key Gap: Systems monitor — but don’t act autonomously

SOLUTION OVERVIEW-

AI-Powered Supply Chain Control Tower

We built a multi-agent AI system that:

- Continuously monitors supply chain operations
- Detects failures in real time
- Uses GenAI to make intelligent decisions
- Automatically executes corrective actions
- Maintains full audit logs

CORE INNOVATION-

Multi-Agent Autonomous Decision System

Our system is NOT a chatbot ,It is a closed-loop AI system:

Detect → Decide → Act → Log → Repeat

Agents:

- Order Agent
- Failure Detection Agent
- Decision Agent (GenAI)
- Action Agent
- Audit Agent

Each agent has a clear responsibility
→ modular + scalable

GEN AI INTEGRATION-

Intelligent Decision Making with GenAI

- Integrated with Hugging Face API
- Prompt-based decision generation
- Extracts structured actions from responses

Example Decisions:

- Reorder stock
- Switch supplier
- Escalate issue
- Mark priority

Fallback logic ensures reliability even if AI fails

SYSTEM WORKFLOW-

End-to-End Supply Chain Flow

Admin → Product → Supplier → Manager → Order → AI Monitoring

What Happens:

1. Order is placed
2. System monitors continuously
3. Failure detected (delay/stuck)
4. AI suggests decision
5. Action executed automatically
6. Logged for traceability

KEY FEATURES

- Real-time AI monitoring
- Autonomous decision execution
- Live dashboard with insights
- Alert system for critical issues
- Explainable AI reasoning panel
- Audit logs for every action

COMPLIANCE & RELIABILITY

Built with Guardrails

- Rule-based fallback system
- Decision validation before execution
- Full audit trail (who/what/when)

Ensures:

- Safe AI behavior
- Transparency
- Production readiness

BUSINESS IMPACT

- Reduces operational delays
- Prevents supply chain disruptions
- Improves decision speed
- Enables real-time optimization
- Scalable for enterprise systems

Why This Stands Out

- Autonomous AI (not just analytics)
- Multi-agent architecture
- Closed-loop intelligent system
- Explainable + auditable decisions

We transformed supply chains from reactive systems into intelligent, self-operating ecosystems.”